

Project Asteria - Engineering Log Blueprint

Vision

Project Asteria will be developed like a real engineering and research project. Every session will generate an Engineering Log documenting discoveries, architecture decisions, trade-offs, concepts learned, and future work.

Engineering Log Structure

1. Mission of the Session
2. What We Knew Before
3. Discoveries (ideas reached during discussion)
4. Architecture Decisions with reasoning
5. Design Discussions and trade-offs
6. Open Questions for future sessions
7. Robotics Concepts Learned
8. Engineering Reflection

Project Repository Structure

Project Asteria

- Engineering Logs
- Architecture diagrams
- Mathematics (kinematics, control, estimation)
- Simulations
- Python Implementation
- Experiments
- Future Ideas

Development Phases

Phase I: Genesis (Architecture)

Phase II: Locomotion (Movement)

Phase III: Perception

Phase IV: Intelligence

Phase V: Exploration

Phase VI: Asteria-X (Research-grade autonomous robot)

Guiding Principle

Never optimize for speed; optimize for understanding. Every architectural decision should have a clear justification so the system can be defended in research discussions and interviews.